

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

7th Semester of B. Pharm. Examination

University Theory Examination April 2016

PH403 Medicinal Chemistry- III

Date: 18.04.2016 , Monday

Time: 1.30 p.m to 4.30 p.m

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following:

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| A Define diuretics. Classify them with examples. Give the synthesis of furosamide and chlorothiazide. | 10 |
| B Define QSAR. Compare the SAR and QSAR. Discuss Free- Wilson analysis. | 10 |
| C Classify various descriptors used for deriving QSAR equation and discuss any one class in detail. | 10 |

Q 2 Attempt any TWO of the following:

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|---|----|
| A What do you mean by lead compound? Discuss various methods for lead optimization. | 10 |
| B Add a short note on: 1) High-Throughput Screening | 05 |
| 2) SOSA Approach | 05 |
| C Add a short note on: 1) Cardiotonics | 05 |
| 2) Antiarrhythmic agents | 05 |

SECTION- II

Q 3 Attempt any TWO of the following:

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|--|----|
| A Add a short note on: 1) Craig's Plot | 05 |
| 2) COMFA | 05 |
| B Add a short note on: 1) Hansch Analysis | 05 |
| 2) Pros and cons of 3D QSAR over 2D QSAR | 05 |
| C Give detailed classification of antihypertensive agents and explain SAR of Calcium Channel Blockers. | 10 |

Q 4 Attempt any TWO of the following:

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| A | Add a short note on: 1) Ligand Based Drug Design | 05 |
| | 2) De- Novo Drug Design | 05 |
| B | Classify antihyperlipidemic agents and give SAR of HMG-CoA reductase inhibitors. | 10 |
| C | Add a short note on: 1) Anti-anginal agents | 05 |
| | 2) SAR of Thiazide diuretics | 05 |
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